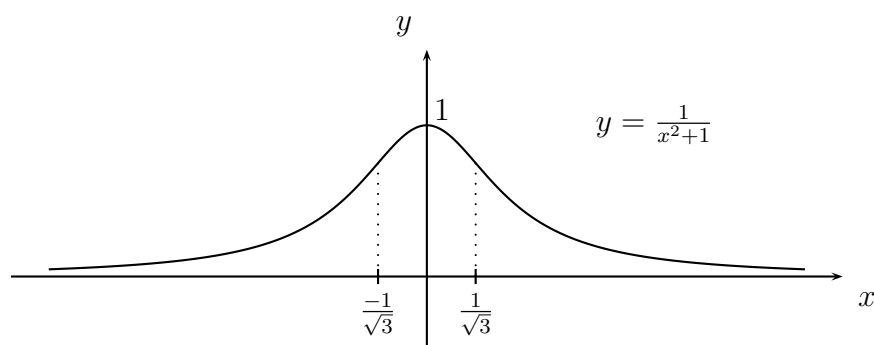


Notes for “An application of the Quotient Rule”

Important Ideas and Useful Facts:

- (i) The witch of Maria Agnesi: The curve $y = \frac{1}{x^2+1}$ is known as the *witch of Maria Agnesi*.



It attains a global maximum 1 at the y -intercept and has points of inflection at $x = \pm \frac{1}{\sqrt{3}}$.

- (ii) **Derivative of the inverse tangent function:** The derivative of the inverse tangent function is the rule that gives rise to the witch of Maria Agnesi, that is,

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{x^2 + 1}.$$

Derivations:

1. Consider the curve $y = \frac{1}{x^2+1} = (x^2 + 1)^{-1}$. The y -intercept is $y(0) = \frac{1}{0^2+1} = 1$. There are no x -intercepts as the numerator is never zero. The derivative, by the Chain Rule, is

$$y' = -(x^2 + 1)^{-2}(2x) = \frac{-2x}{(x^2 + 1)^2},$$

with the following sign diagram:

x		0	
y'	$+$	0	$-$

This indicates that the curve is increasing for $x < 0$ and decreasing with $x > 0$, attaining a global maximum of $y = 1$ when $x = 0$.

To find the second derivative, by the Quotient Rule, put $u = -2x$ and $v = (x^2 + 1)^2$, so that $u' = -2$ and, by the Chain Rule,

$$v' = 2(x^2 + 1)(2x) = 4x(x^2 + 1).$$

Cancelling $(x^2 + 1)$ in the numerator and denominator at the third step, we get that

$$\begin{aligned} y'' &= \frac{vu' - uv'}{v^2} = \frac{(x^2 + 1)^2(-2) - (-2x)(4x)(x^2 + 1)}{(x^2 + 1)^4} \\ &= \frac{(x^2 + 1)(-2) + 8x^2}{(x^2 + 1)^3} = \frac{-2x^2 - 2 + 8x^2}{(x^2 + 1)^3} = \frac{6x^2 - 2}{(x^2 + 1)^3} \\ &= \frac{2(3x^2 - 1)}{(x^2 + 1)^3}, \end{aligned}$$

which is zero precisely when $3x^2 - 1 = 0$, that is, $x = \pm \frac{1}{\sqrt{3}}$. We get the following sign diagram for y'' :

x	$\frac{-1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$			
y''	+	0	-	0	+
	⋘		⋙		⋘

This indicates that the curve is concave up for $x < \frac{-1}{\sqrt{3}}$ and for $x > \frac{1}{\sqrt{3}}$ and concave down for $\frac{-1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, with inflections where the concavity changes at $x = \pm \frac{1}{\sqrt{3}}$.

This completes the verification of all of the main features of the curve known as the *witch of Maria Agnesi*.

2. We now explain why the derivative on the inverse tangent function gives the rule that describes the witch of Maria Agnesi.

We start by recalling the circular identity

$$\sin^2 x + \cos^2 x = 1,$$

and divide both sides by $\cos^2 x$, to get

$$\left(\frac{\sin x}{\cos x}\right)^2 + 1 = \frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \left(\frac{1}{\cos x}\right)^2.$$

Recall that $\tan x = \frac{\sin x}{\cos x}$ and $\sec x = \frac{1}{\cos x}$, so this becomes another well-known identity in trigonometry:

$$\tan^2 x + 1 = \sec^2 x.$$

Put $y = \tan x$, so that $x = \tan^{-1} y$. Hence, using this identity, we get

$$\frac{dy}{dx} = y' = \sec^2 x = \tan^2 x + 1 = y^2 + 1,$$

so that, reciprocating,

$$\frac{d}{dy}(\tan^{-1}(y)) = \frac{dx}{dy} = \frac{1}{y^2 + 1}.$$

Using x as usual as the input variable, we get the required result:

$$\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{x^2 + 1}.$$